Prometheus Rocket Project

Structural Components, Fins, and Nosecone

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ABSTRACT

The Prometheus Rocket Project was Walla Walla University's third iteration of a multistage rocket project. This year's design expanded previous work by incorporating Level-2 motors and an active control fin system. These additions required a larger rocket body, stronger structural components, increased control surface area, and more complex internal systems. The added space also allowed for larger electronics bays, improved recovery components, and an onboard camera for flight footage. Throughout the design process, the team applied engineering principles from coursework in structures, aerodynamics, materials, manufacturing, and controls to develop a rocket capable of supporting future aerospace projects at Walla Walla University.

INTRODUCTION

For a successful launch, the body tubes, nosecone and fins must be sized and built appropriately to ensure stable flight without breaking apart. The body tubes need to be strong enough to sustain the acceleration and the maximum dynamic pressure of liftoff. In addition, the tubes must withstand the force of the ejection charges encountered during stage separation and recovery. The fins are essential for stability, as they significantly impact the center of pressure location on the rocket. The second stage fins must be compatible with the active control system attached. This means that a servo must be able to fit inside of the fin. The nosecone is also essential for stability, as its mass drives the center of mass forward which increases stability.

DESIGN PROCEDURE

Body Tubes

To make the main body tubes strong and lightweight while maintaining a low budget, a fiberglass-epoxy composite was selected. The primary forces that the tubes encounter are the thrust from the motor, velocity of the rocket, and the maximum dynamic pressure inside the tube after launch. These forces were converted into global hoop and axial stresses to determine the stress encountered by each ply. Composites are anisotropic, meaning the calculations are more complex. Tsai-Wu Failure Theory was used to determine what grain orientation should be used as well as the number of plies. The result was a 6-layer composite with a $[0,45]_3$ which yielded a factor of safety of 3.76

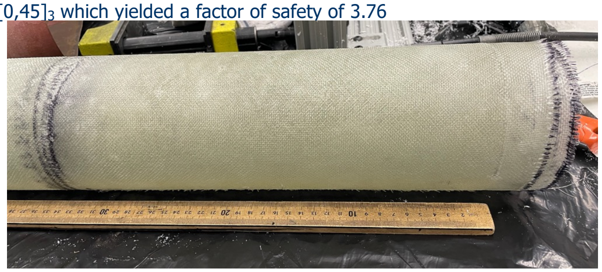

Figure 1: Body Tube

DESIGN PROCEDURE CONTD.

First Stage Fins

The first-stage fin system uses passive control, meaning the fins remain fixed throughout flight. The fins were manufactured from the same fiberglass-epoxy composite as the fuselage, which simplified material compatibility while maintaining a high strength-to-weight ratio. Each fin consists of 16 alternating lamina, producing an average thickness of 0.425 cm. A fin jig was used to attach the fins at 90° spacing around the body tube. Using machine design principles, a fiberglass-reinforced epoxy fillet with a 1.5 cm radius was calculated to provide sufficient strength against thrust loads, drag forces, and flutter deflection.

Second Stage Fins

The primary concerns when designing fins for a rocket are stability and aerodynamics. The ideal fin is as thin, rigid, light and moderately sized. The active control system requires that a servo, bearings, rod, and fin tab be attached to this fin. This greatly increased the thickness and weight of the fin which impacted aerodynamics as well as optimal size. To account for the complex geometry, the fins were 3D printed at 30% infill, and were strengthened with 3 layers of fiberglass on the outside. The fins were an airfoil shape to reduce drag as much as possible.

Nosecone

A critical part for the aerodynamics of all rockets is the nosecone. The Prometheus Rocket was predicted to travel at Mach 0.9 meaning it will be in the transonic region. This means that wave drag was a significant portion of the drag experienced by the device. To reduce this, a longer nosecone was preferred to reduce the forming of this wave. The nosecone must also be structurally stable enough to survive launch forces. Balancing these two factors lead to an optimal fineness ratio of 5.8:1 for length of nosecone to internal diameter of the rocket.

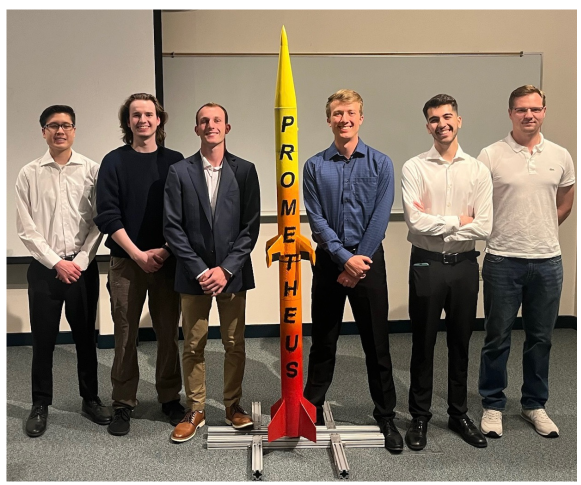

Figure 2: Prometheus Team with Rocket

DESIGN PROCEDURE CONTD.

2nd Stage Electronics Bay

For the electronics and control system to work properly, the boards must be securely attached to the rocket without exposure to vibration. To ensure these criteria are met, a threaded rod was secured between bulkheads, and the electronics bay was mounted to this rod. The electronics bay was made of PETG filament for high strength and durability and included large fillets to prevent failure during launch. Brass threaded heat sinks were inserted into the electronics bay to ensure secure attachment of the boards. To reduce vibration to the boards, a nylon washer was inserted on either side of the board where screwed in.

VALIDATION

To validate calculations, a test strip was manufactured and compression tested to failure. This provided a way to verify manufacturing technique as well as confirm results. The test strip failed at 2750lb which greatly exceeded expectations.

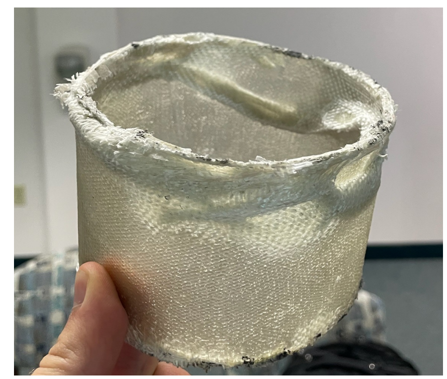

Figure 3: Composite Test Strip

SUMMARY

In summary the team successfully demonstrated aerospace engineering principles through the application of course work concepts learned at WWU. The team looks forward to future iterations and are excited to see what will be added next.

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[1] Zavorotny, Andrew, Anthony Lin, Anthony Talevi, David Neptune, Ethan Chan, Jason Killen, Marc Daniels, et al. 2018. "Team 55 Project Technical Report for the 2018 Spaceport America Cup." University of Ottawa Student Team of Aeronautics and Rocketry.

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Prometheus Rocket Project

The Guts of Prometheus

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ABSTRACT

With the goals of Prometheus, the internals needed to be able to withstand high powered rocket motors. The internals were designed and built using a variety of methods. This process included designing and building components such as the mechanical control system, the motor mounts and motor retention system, the bulkheads and couplers, and the electronics bays. Unfortunately, the rocket has not flown, primarily due to issues with the couplers. Since the rocket did not launch, data has been collected from OpenRocket simulations. Additionally, a successful assembly shows that the internals function as intended without flight.

INTRODUCTION

While the body of Prometheus is what is seen from the outside, the internals are what cause the rocket to function. This section of the Prometheus Rocket Project Poster Set covers the physical internal portions of Prometheus, aside from the electronics boards. These internals needed to be capable of holding all the external pieces of the rocket together, the force required to lift the rocket, and the equipment to bring the rocket down safely. Additionally, this poster covers the manufacturing used on the pieces of Prometheus.

DESIGN PROCEDURE

Motors

Before design could be begin, the initial choices for motors had to be selected. This would give predictions such as thrust, speed, and height for the team to base designs on. The motor for each stage had different goals. The first stage needed to provide enough thrust to lift the rocket and propel it to max velocity. The second stage motor needed to carry the speed and provide enough flight time for the active control system to be showcased. An L1000 motor was chosen for the first stage, it provides a large amount of thrust that will bring the rocket up to speed. The second stage motor, a J435, provides enough thrust to propel the second stage, while not providing more thrust than the active control system can handle. To abide by regulations set by the National Association of Rocketry, these motors were purchased, not made [2].

Recovery

Designing and manufacturing the recovery system involved several different aspects coming together. The decision was made to use a dual deploy system to ease in the recovery process and increase the chances of a successful flight. Sizing of parachutes, ejecting charges, and the dual deploy bay itself were the primary portions of the design. Testing was competed on the ground to ensure enough powder was used to fully eject the chute.

DESIGN PROCEDURE CONTD.

Motor Mounts

To contain the motors in the rocket, motor mounts were designed and manufactured. Both motor mounts consisted of motor tubes, centering rings, and motor retention caps. Motor tubes connected the motors to the centering rings, which then connected to mounts to the rocket's body tube. Motor retention caps are in place at the end of the motor mounts to keep the motors from falling out of the rocket. The lowest centering ring on the second stage has been modified to accommodate for the servos used in the active control system. All parts of the motor mounts, aside from the motor tubes, were 3D printed using PETG. Analysis was done on the mounts in ANSYS to ensure their reliability during launch.

Active Control System

Two stage rockets commonly run into issues staying vertical after stage separation. To attempt to combat this, an active control system was built into the second stage fins. This control system works using four servos that turn a fin tab in each fin. These tabs will be adjusted through the flight to control the rocket in pitch and yaw. Layout for this system was a challenge due to the size constraints and the force that needed to be overcome by the fin tabs. This system was tested along side the rest of the systems on the ground before launch repeddle.

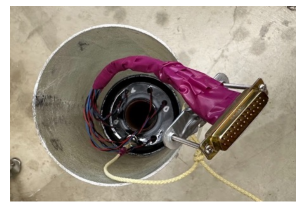

Figure 2:Centering Rings

Figure 3: Assembled Control System

INTERNAL LAYOUT

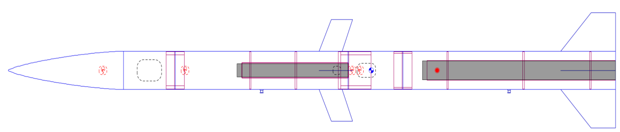

Figure 4: Internal Layout Diagram

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[1]"Discord Server!," *Discord*,2026. https://discord.gg/SBg9WeKc (accessed May 04, 2026).

[2]National Association of Realtors. n.d. "Laws and Regulations." Accessed May 26, 2026. National Association of Realtors Laws and Regulations

DESIGN PROCEDURE CONTD.

Couplers and Electronics Bays

The couplers' function was to hold the lengths of body tube together and allow for separation between parts of the rocket. In total there were four different coupler pieces needed to hold 5 different tube sections together. These couplers each were slightly different due to the differences in function each served. The couplers were paired with bulkheads in several places to allow pressure barriers. Several of the couplers were also built into electronics bays. This provided an easy way to access and charge the electronics and protected them from outside interference. Unfortunately, the neck length on the designed couplers was deemed too short to safely fly. The recommended length was 1-1.5 cal in length

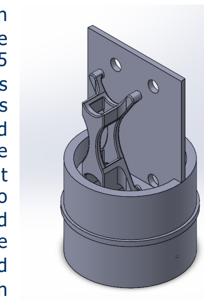

Figure 5: 1st Stage Electronics Bay

RESULTS

On the intended day of launch, there were several issues with the rocket that led to the rocket not being launched. The main reason for this was the coupler shoulder lengths. They were not long enough to ensure a safe flight without risk of buckling at the coupler connections. All other internal systems including the recovery and the electronics bays were verified as adequate before launch.

However, from the OpenRocket simulations, projected values for flight data were collected. By setting the timings for the flight path, the height and velocities were maximized. Choosing stage separation, via hot staging, to occur 7 seconds after the booster motor burns out, the velocity of the second stage is preserved. Then, using the maximum delay possible on the sustainer motor, the apogee is reached before the drogue chute is deployed. The maximum velocity, apogee, and acceleration can be seen in the figure below.

Flight configuration:	The One
Apogee:	3223 m
Max. velocity:	279 m/s (Mach 0.824)
Max. acceleration:	173 m/s²

Figure 6: Data Outputs from OpenRocket

Summary

The internals of the Prometheus rocket project hold the entire rocket together. This was accomplished using many design, manufacturing and assembly methods. Unfortunately, this rocket was not flown, however what was made acts as an excellent proof of concept and example of engineering expertise.

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Prometheus Rocket Project

Flight Computer & Active Stabilization

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ABSTRACT

Given the objectives of the Prometheus Rocket, hardware/software was needed to facilitate altitude-based parachute deployment, active stabilization via a control system, and data logging. To accomplish this, the Pronoia flight computer was designed as a 4-layer PCBA which included power management, actuation for rocket motor ignition and pyro-channel ignition, and a microcontroller socket. For the microcontroller, a Teensy 4.1 was chosen for the following features: a 600 MHz clock rate, a Floating Point Unit (FPU), a solderable flash chip footprint communicating over QSPI, and its arduino-compatibility. For embedded software development, PlatformIO via VSCode was used as opposed to the Arduino IDE due to its enhanced project management and compatibility with other VSCode extensions.

INTRODUCTION

With the advancement of amateur rocketry and the Prometheus Rocket being the 3rd rocketry iteration at WWU, an embedded system is not only preferred but necessary to accurately accomplish objectives. This decision to move past Commercial-Off-The-Shelf (COTS) products necessitates a thorough Bill Of Materials (BOM) including all electronic components, a power management subsystem, PCBA schematic design, PCBA layout, embedded software development, LTSpice simulation of electronic behaviors, and software/control system simulation via RocketPy.

DESIGN PROCEDURE

During the development phase of the project, system requirements were defined for the performance of the project, and parts were chosen conservatively according to their specifications. This design strategy was applied across all subsystems, including power management, flight computer components, the pre-flight protocol, software, and control.

Power Management

To actively-stabilize the rocket, controllable fin tabs were embedded into the rocket's canards (or second-stage fins). These fin tabs were controlled via servo motors. The servo motors chosen had a high torque capability but required a high voltage and a high-current in the case of a servo motor stall. So, the flight computer intakes a 95C 2S LiPo battery at its maximum voltage of 8.4 V. Then, the 8.4 V trace gets run to the components that utilize high power, including the servo motors, pyro-channels, and recovery buzzer. Furthermore, the 8.4 V runs through a 5V buck converter, which powers the Teensy 4.1. To reduce EMI and cross-talk, Power Management and high-current components are partitioned to the left of the PCBA, while 3.3 V logic level components are partitioned to the right, as seen in Figure 1. To serve the high-current draw needs of the pyro channels and servo motors, two bulk capacitors are placed near them, respectively.

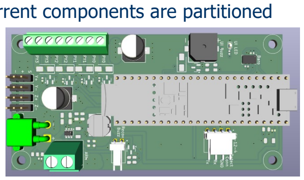

Figure 1: Flight Computer Front-Side

DESIGN PROCEDURE CONTD.

Flight Computer Components

The main flight sensor suite includes an ICM 42688 IMU (±16 G Accel and ±2048 dps), a KX134 High-G Accelerometer (±64 G), and a MS5611 Barometer. These high-performance components were chosen to replicate other advanced college rocketry projects, such as MIT's Pyxida Flight Computer [1]. The addition of a High-G accelerometer is necessary as the Prometheus rocket experiences over 16 G's, while the IMU is still needed for its precision and integrated gyroscope. 4 pyro-channels were included to account for second stage ignition, main parachute deployment, and 2 extra in the case that a drogue parachute trigger and a stage separation trigger are needed. Each pyro-channel, along with the UI Buzzer and Recovery Buzzer, which all necessitated a higher current draw than the Teensy could supply, were designed with a N-Channel MOSFET driver, so that their power source comes directly from the 2S LiPo battery instead of the microcontroller.

Pre-Flight Protocol

Given the complexity and L2 status of the Prometheus Rocket project, safety was a primary concern. Thus, a robust pre-flight protocol is designed to monitor the rocket and complete a variety of tasks. These include but are not limited to battery voltage monitoring, barometer accelerometer and gyroscope calibration, switching power on to pyro channels and servo motors, sensory access to a UI buzzer/LED, ensure data logging, software arming the rocket, ensuring pyro-channel continuity, etc. These tasks are accessible while the rocket is set up on the rail, via a wired GX-12-4 UART connection between the flight computer and a laptop, running a Command Line Interface (CLI). This ensures safety and performance, as it enables necessary tasks to be completed before lift-off with the rocket is on the rail.

Software

The beginning of embedded software development always starts with hardware bring-up. The sensor suite, flash chip, and GX-12-4 connection had to be configured and interfaced with. Next, a flight state machine was designed for both the 1st and 2nd stages, which tracks the current flight phase each stage is currently in. Strategy and code was referenced from SparkyVT's open-source GitHub repository [2].

SYSTEM BLOCK DIAGRAM

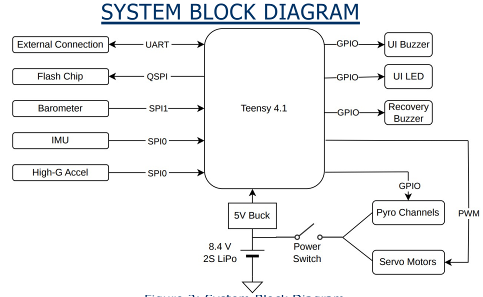

Figure 2: System Block Diagram

DESIGN PROCEDURE CONTD.

Control System

To accurately achieve rocket stabilization via fin tabs, a control system must compare a setpoint with its estimated current state to command an action based on the rocket's dynamics. The Prometheus Rocket's setpoint is a degree-per-second (dps) value of zero, across all 3 rotational axes (roll, pitch, and yaw). The commanded actions are commanded servo angles, which translates to an applied torque on the rocket.

The cornerstone of this control system is the accuracy of its estimated inertial state. In particular, the control system and Prometheus objectives rely on accurate speed estimation and accurate altitude estimation. Speed estimation is required for calculating the aerodynamic force on a control surface, which informs a correct fin tab angle to produce a desired torque. Altitude estimation is required for accurate altitude-based parachute deployment and second stage motor ignition. For altitude estimation, a Complimentary Filter is used to combine the barometric altitude reading and the double-integrated acceleration [2].

Proportional Derivative (PD) Control System Block Diagram

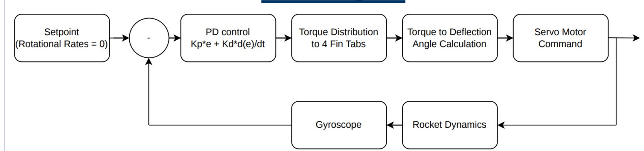

Figure 3: Control System Block Diagram

Results and Summary

The goal of the Prometheus Rocket avionics stack was to implement an active control system for rocket stabilization in all 3 rotational axes, log flight data via a camera and a flash chip, and trigger the second stage motor and parachutes via pyro channels. Although these objectives were not yet verified in flight, the resources and knowledge gathered for the next rocketry group will prove beneficial for WWU. These include two Teensy 4.1s, five PCBA flight computers, and a full codebase and KiCAD project at *https://github.com/nguyjo/Project_Prometheus*. Accomplishing these goals came about largely due to the pursuit of simplicity and finding the minimum solution. By substituting GPS and telemetry with a recovery buzzer, and Bluetooth communication for a wired pre-flight protocol, the overhead was largely reduced. However, the values of component redundancy, testing, and simulation, across multiple design iterations, were not displayed during the Prometheus rocket project due to financial constraints, time constraints, and the large distance between WWU and the Oregon rocketry site.

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[1]"Pyxida - MIT Rocket Team - MIT Wiki Service," Mit.edu, 2020. https://wikis.mit.edu/confluence/display/RocketTeam/Pyxida (accessed May 06, 2026).

[2]SparkyVT, "HPR-Rocket-Flight-Computer," GitHub. https://github.com/SparkyVT/HPR-Rocket-Flight-Computer (accessed May 05, 2026).

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